

flyPad landing calculator (A380 new, A320 reworked)

The A380 OIS LDG PERF application on the flyPad of both aircraft, with FCOM data, FMS import and BTV exit advice

Branch	pr/23-efb-landing-calculator
Aircraft	A380X and A32NX (flyPad, A380 FMC, A320 MCDU)
Commits	fc850a71e feat(efb): A380 landing calculator and reworked A320 landing calculator; e7e27cf62 fix(efb): strict null checks of the landing calculator
Base	pr/12-a320-takeoff-calculator (shared performance module and FMS links of the takeoff calculators)
Status	CI checks pass; merged into develop 224cf6741. Checked in the flyPad harness; BTV exits seen in the sim once; full in-sim test to do.

Purpose

The A380X flyPad had no landing calculator and the A32NX one had only the QRH in-flight distances, no dispatch data, no FMS import and no approach or go-around parameters. The real A380 computes landing performance in the OIS LDG PERF application (A380 FCOM PER-LND).

Specification (A380 FCOM PER-LND, DSC-32-10-30-20, LIM-32-30; A320 FCOM PER-LDG-DIS, QRH)

Part	Content
Computation	DISPATCH (factored required landing distance) or IN-FLIGHT (actual landing distance)
Runway	airport, runway, LDA, heading, elevation, slope; A380: go-around altitude
Conditions	runway condition (dry, wet, RWYCC contaminants), wind, temperature, QNH; A380: anti-ice, air conditioning
Aircraft	landing weight in t / klb with an MLW button; CONF (AUTO on the A380), go-around CONF and gradient
Landing	approach type, VLS+ increment, autoland and glide slope, braking mode (manual, autobrake, BTV on the A380), reverse thrust (greyed out without credit); A320: overweight procedure
RESULTS	LW, MLW(PERF) and its limitation (WGT / LDA / ACG), flaps, landing distance, stop margin, go-around speed and gradient, VAPP = VLS + VLS+; estimates in amber
Runway picture	touchdown, the stop point of each braking mode, the required distance; A380: BTV DRY / WET lines and the runway exits

A380 data and rules

- Landing field length chart of the Airbus A380 Aircraft Characteristics (dry, manual braking, CONF FULL, VLS), extracted from the vectors; a model fitted on it (air time 5.149 s, dry deceleration 3.253 m/s², 0.8 % RMS) for the other conditions.
- Dispatch: RLD = ALD / 0.6, wet x 1.15, contaminated = max(wet RLD, ALD x 1.15); 50 % of the headwind, 150 % of the tailwind; temperature only for water-contaminated runways or autoland; slope only for autoland (PER-LND).
- Autobrake decelerations LO / 2 / 3 / HI = 2.0 / 2.5 / 3.0 / 3.5 m/s² (DSC-32-10-30-20).
- Go-around gradient = T/W - K, calibrated on the FCOM in-flight example (386 t: 8.939 %; MLW(PERF) 569 t at the 2.7 % minimum; GA speed 141 kt).

BTV exit advice (A380, information only)

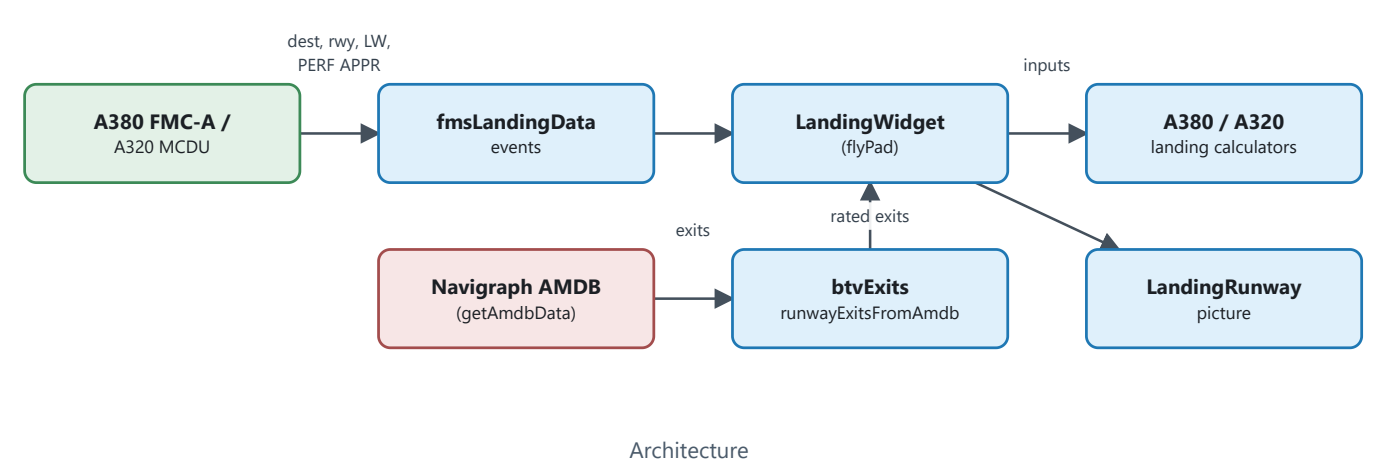
DRY / WET lines = 400 m touchdown + 5 s roll + $V^2 / (2 \times 2.8 \text{ m/s}^2 \text{ dry, } 1.8 \text{ m/s}^2 \text{ wet})$, V = TAS at VAPP minus the wind (FCOM DSC-32-10-30-20 and the FBW autobrake constants). The runway exits of the Navigraph airport map (OANS

exit rules: within 50 m of the centreline, at least 400 m after the threshold, turn up to 120 degrees) are rated: recommended (first exit beyond WET), beyond WET, dry runway only, not achievable. Nothing is shown on a contaminated runway (BTV prohibited, LIM-32-30).

A320 data

- QRH in-flight landing distances kept (x 1.15 factored shown).
- FCOM dispatch required landing distance tables (PER-LDG-DIS-RLD), autoland correction, wet minimum.
- Crosswind limits of PER-LDG-DIS-MAT; reverser credit per QRH correction.

Component added or changed by the feature MSFS / simulator Aircraft system (systems host, FMS, FWS) Display / page
Data, LVar, storage Crew, flypad, external program



Interfaces

Name	Type	Meaning
fms_landing_data_request	event bus	the flyPad asks the FMS for its landing data (request id)
fms_landing_data	event bus	destination, runway, landing weight, QNH, temperature, wind, CONF

Tests

16 unit tests: `a380x_landing.spec.ts` (8), `a32nx_landing.spec.ts` (6), `btvExits.spec.ts` (2).

Limitations

- Only the A380 dry manual-braking chart is Airbus data; every other A380 value is a model estimate (amber).
- The BTV exits need a Navigraph airport map; the advice does not arm the BTV (that stays on the OANS).

PR description (from the PR draft)

From `pr-drafts/23-efb-landing-calculator.md`, section "Summary of Changes" onwards.

Summary of Changes

One landing screen for both aircraft, laid out after the A380 OIS LDG PERF application (A380 FCOM PER-LND): a DISPATCH / IN-FLIGHT computation type, Runway / Conditions / Aircraft / Landing sections, a RESULTS panel and a runway picture.

- **Inputs:** airport and runway (LDA, heading, elevation, slope), runway condition (dry, wet and the RWYCC contaminants), wind, temperature and QNH, anti-ice and air conditioning (A380), landing weight in tonnes (or klb) with an MLW button, CONF (AUTO on the A380), go-around CONF and gradient, approach type and VLS+ increment, autoland and glide slope, braking mode (manual, autobrake modes, BTV on the A380), reverse thrust (greyed out when the condition gives no reverser credit), overweight procedure (A320).
- **RESULTS:** landing weight, MLW(PERF) and its limitation code (weight, landing distance, go-around gradient), flaps, landing distance (factored for dispatch), stop margin, go-around speed and gradient, VAPP = VLS + VLS+; estimated values in amber.
- **Runway picture:** touchdown, the stop point of each braking mode, the required distance bar, and on the A380 the BTV DRY / WET lines with the runway exits.
- **A380 data:** the landing field length chart of the Airbus A380 AC document (dry, manual braking, CONF FULL, VLS) and a model fitted on it (0.8 % RMS) for the other conditions. FCOM rules: dispatch RLD = ALD / 0.6, wet x 1.15, contaminated = max(wet RLD, ALD x 1.15); 50 % of the headwind, 150 % of the tailwind; temperature only for water-contaminated runways or autoland, slope only for autoland; autobrake decelerations LO / 2 / 3 / HI (DSC-32-10-30-20); go-around gradient calibrated on the FCOM in-flight example.
- **BTV exit advice (A380, information only):** DRY / WET lines = 400 m + 5 s roll + the braking distance at the BTV decelerations; the runway exits of the Navigraph airport map (the OANS exit rules) rated as recommended (first exit beyond the WET line), beyond WET, dry runway only, or not achievable. Not shown on contaminated runways (BTV prohibited, LIM-32-30).
- **A320 data:** the QRH in-flight landing distances are kept; the FCOM dispatch required landing distance tables (PER-LDG-DIS-RLD), the autoland correction and the wet minimum are added; the crosswind limits of PER-LDG-DIS-MAT.
- **FMS import:** the A380 FMC-A and the A320 MCDU answer the flyPad (shared `fmsLandingData` events) with the destination, runway, predicted landing weight and the PERF APPR QNH, temperature, wind and landing configuration.
- The old `RunwayVisualizationWidget` is removed.
- 16 unit tests (A380 calculator 8, A320 calculator 6, BTV exits 2).

Cockpit API Changes

None (the FMS data travels on two new event bus topics, `fms_landing_data_request` and `fms_landing_data`).

Screenshots (if necessary)

TO ADD: the A380 IN-FLIGHT and DISPATCH screens with results, the A320 screen, the BTV lines with the exits of an airport.

References

- A380 FCOM PER-LND (landing performance, OIS LDG PERF), DSC-32-10-30-20 (autobrake and BTV), LIM-32-30; Airbus A380 Aircraft Characteristics (landing field length chart).
- A320 FCOM PER-LDG-DIS (dispatch RLD tables, autoland, crosswind), QRH in-flight landing distances.

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. A380 flyPad > Performance > Landing: IN-FLIGHT, fill from the FMS (destination and runway in the F-PLN, PERF APPR filled): runway, weight and conditions are imported. CALCULATE: RESULTS and the runway picture.
2. Change the runway condition, braking mode, CONF AUTO, reverse thrust: the distances and the MLW(PERF) change; reverse thrust is greyed out when it gives no credit.
3. With a Navigraph airport map: the BTV lines and the exits, with their rating; a contaminated runway hides them.
4. DISPATCH: the factored required landing distance.
5. A320 flyPad: the same screen with the A320 data; DISPATCH uses the FCOM tables.
6. `npx vitest run` on `a380x_landing.spec.ts`, `a32nx_landing.spec.ts`, `btvExits.spec.ts` (16 tests).